Radiation-Induced Defect Evolution and Device Degradation in Silicon

Roadmap for full-scale multiphysics modeling to reduced-order prediction

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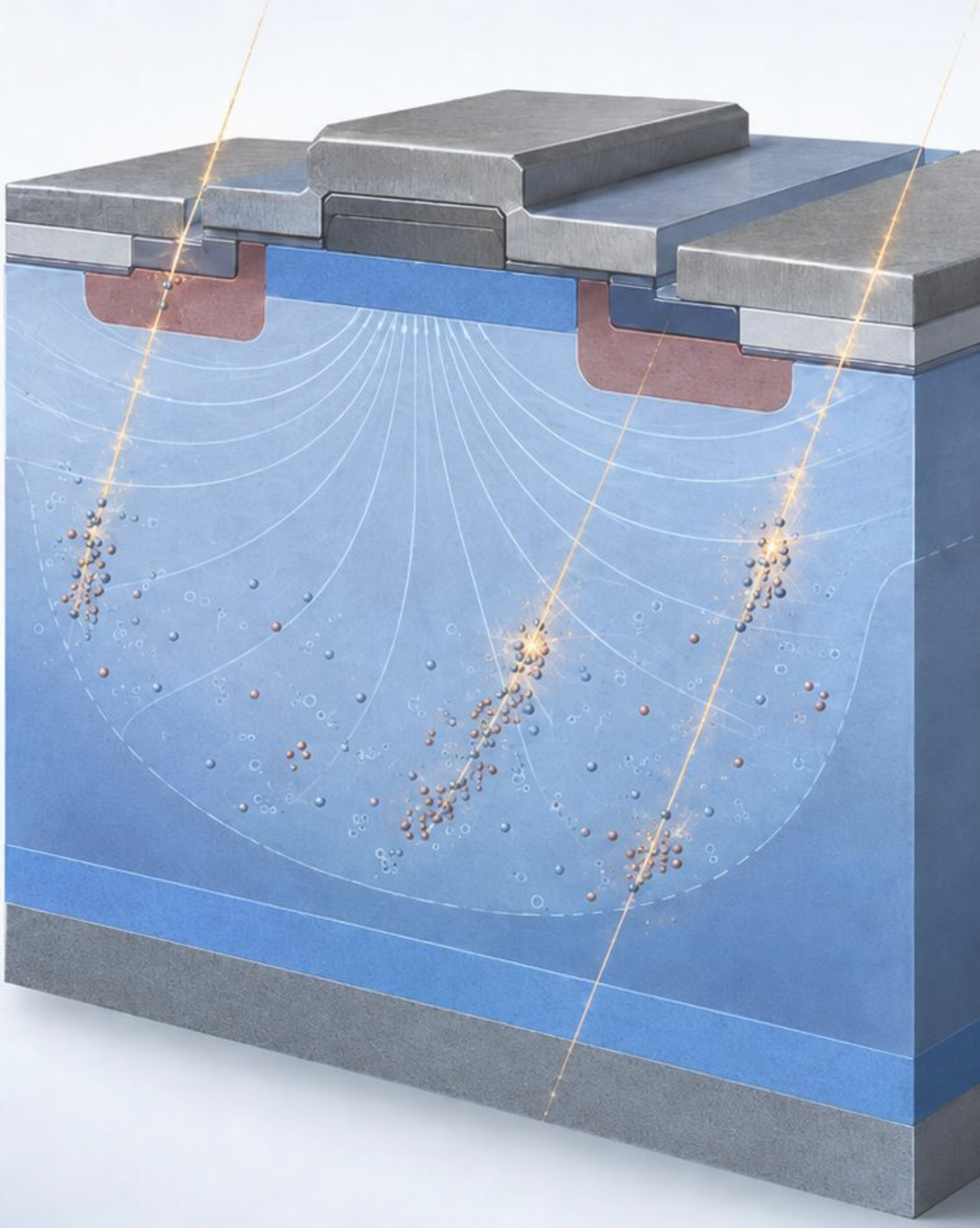
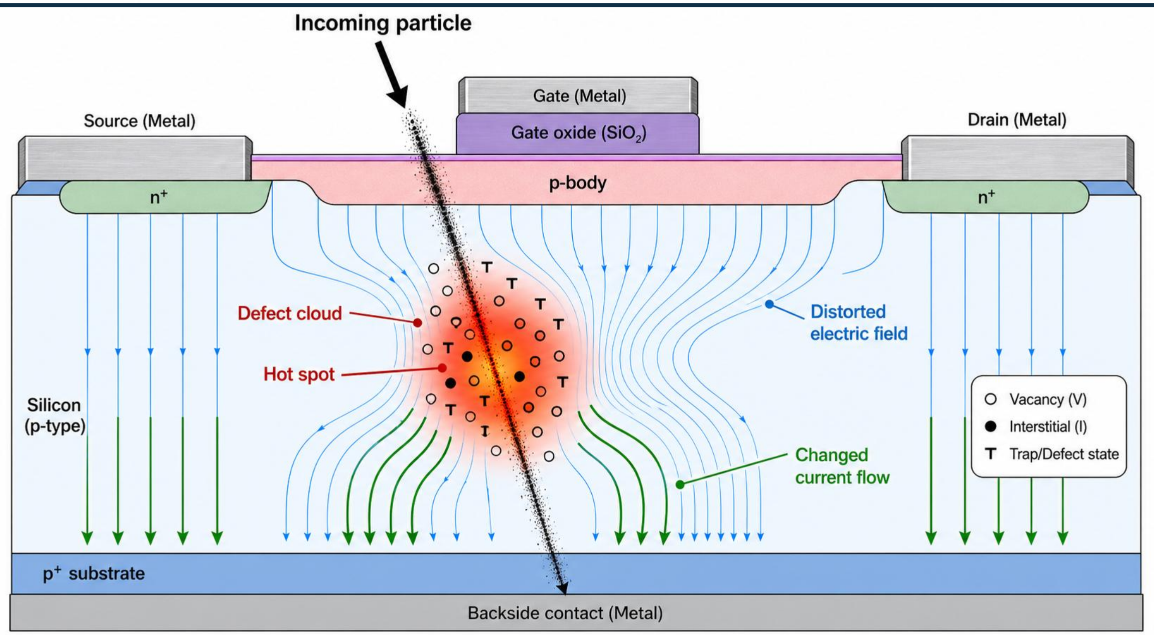
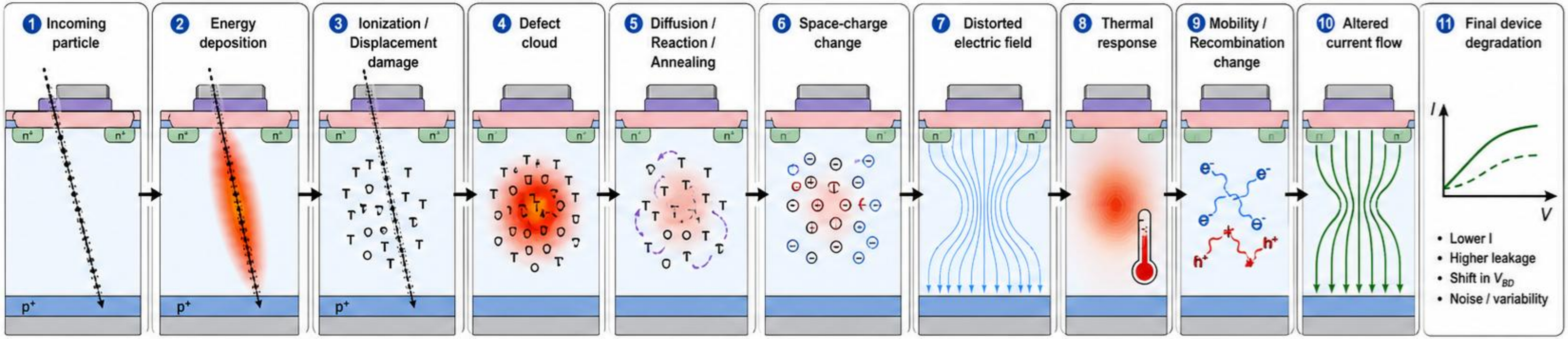


Efficient Radiation-Effects Modeling Requires Preserving Only the Dominant Physics

- Full Multiphysics framework with ∞ time/resources
  - Mod1: Geant4 radiation transport
  - Mod2: Electrostatics
  - Mod3: Defect diffusion and annealing
  - Mod4: Heat transport
  - Mod5: Charge carrier transport
  - Mod6: Multiphysics coupling of Mod2 – Mod5

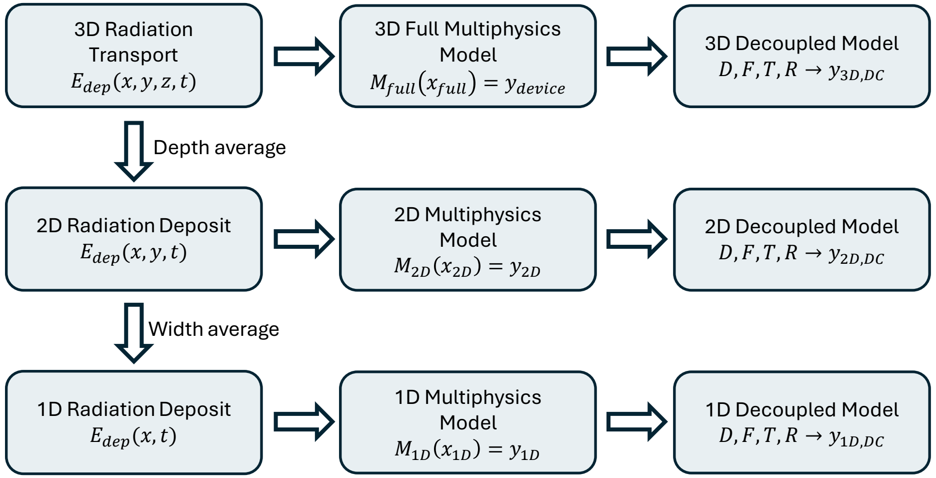
- Reduced complexity model with focused dominant physics.
  - Impact estimation of each module
  - Stepwise reduction of sim with minimum fidelity loss
  - Extrapolation back to full-scale interpretation. (TBD)
  - Uncertainty quantification (TBD)



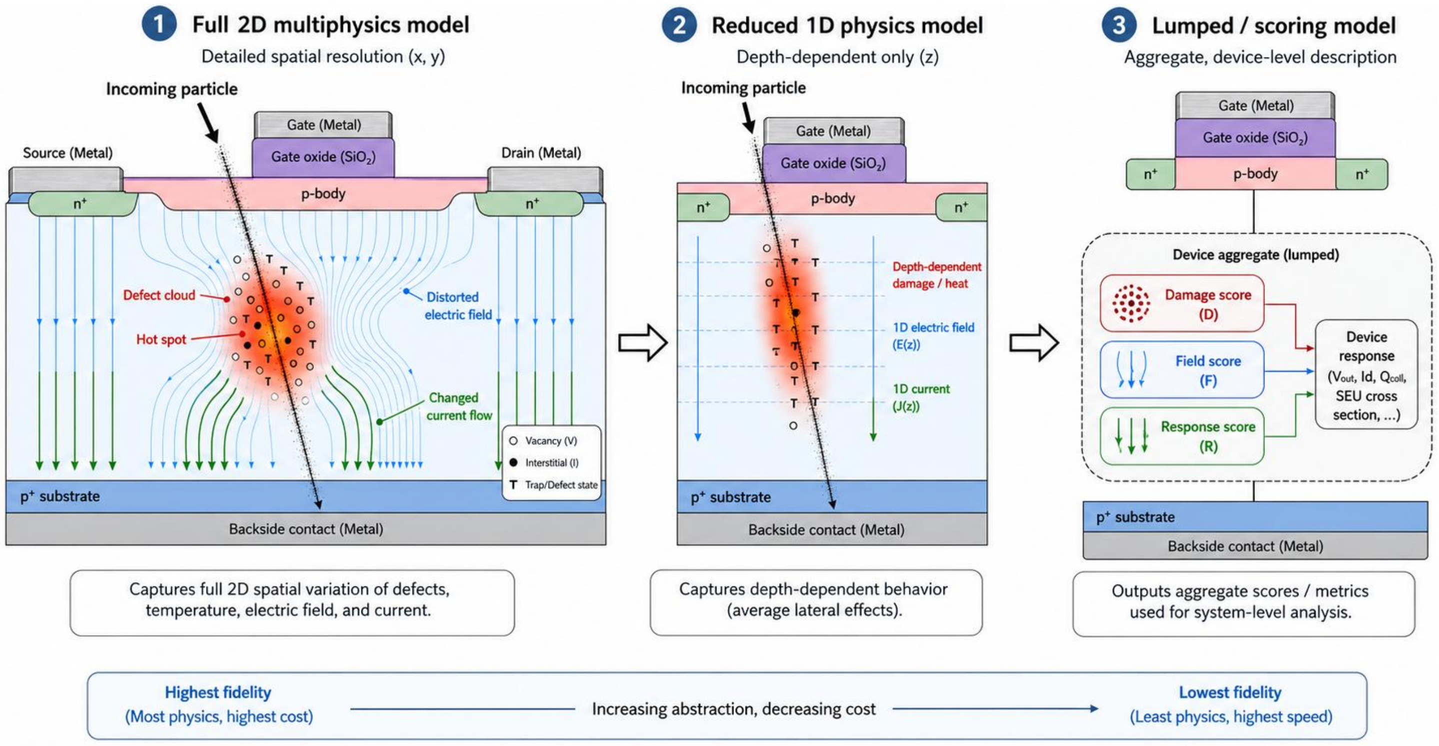


Reduced Dimension and Decoupled Modeling Strategy

- Flowchart of model complexity reduction



$$E_{reduce} = \frac{\|y_{device} - y_{reduce}\|}{\|y_{device}\|} \leq E_{tolerance}$$



- Decoupled model module scoring / Final device response by mapping g (TBD)

$$D_{score}(t) = \frac{\sum_j \int_{\Omega_{2D}} w(\boldsymbol{r}) C_j(\boldsymbol{r},t)\, dA}{\sum_j \int_{\Omega_{2D}} w(\boldsymbol{r})\, dA + \epsilon_0}$$

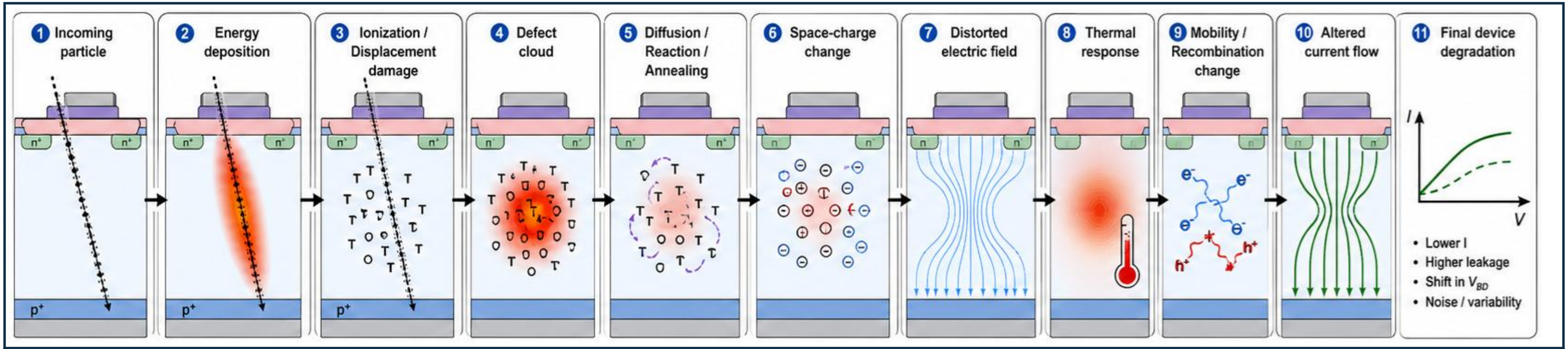
$$T_{score}(t) = \frac{\max_{r\in\Omega_{2D}} |\boldsymbol{T}(\boldsymbol{r},t) - T_0|}{\Delta T_{ref}}$$

$$F_{score}(t) = \frac{\max_{r\in\Omega_{2D}} |\boldsymbol{E}(\boldsymbol{r},t)|}{E_{ref}}$$

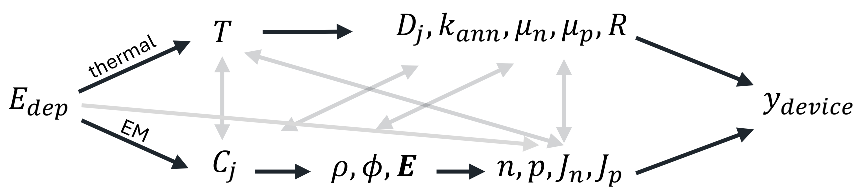
$$R_{score}(t) = g(D_{score}, F_{score}, T_{score})$$

$D_{score}(t)$	Defect damage magnitude at time t (weighted avg)	$w(\boldsymbol{r})$	The weight function acting as device sensitivity map	$\boldsymbol{E}(\boldsymbol{r},t)$	Electric field
$T_{score}(t)$	Thermal contribution to device degradation at time t	$C_j(\boldsymbol{r},t)$	Concentration of defect species j	E_{ref}	Reference electric field
$F_{score}(t)$	Electric field distortion magnitude at time t	ϵ_0	Regularization constant; prevents division by zero	Ω_{2D}	Reduced 2D cross-section domain for sim
$R_{score}(t)$	Final device response to radiation at time t	T_0 / ΔT_{ref}	Reference temperature / Reference temperature rise	E_{dep}	Energy deposition by radiation

Reduced Dominant-Path Model



- Physical sequence: mutually coupled, feeding back into earlier stages

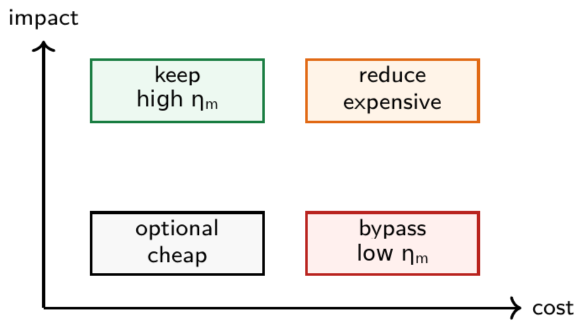


- Current computational coupling sequence (In Progress)

$$E_{dep} \longrightarrow C_j, D_j, k_{ann} \longrightarrow \rho, \phi, \boldsymbol{E} \longrightarrow \mu_n, \mu_p, R, n, p, J_n, J_p \longrightarrow T \longrightarrow y_{device}$$

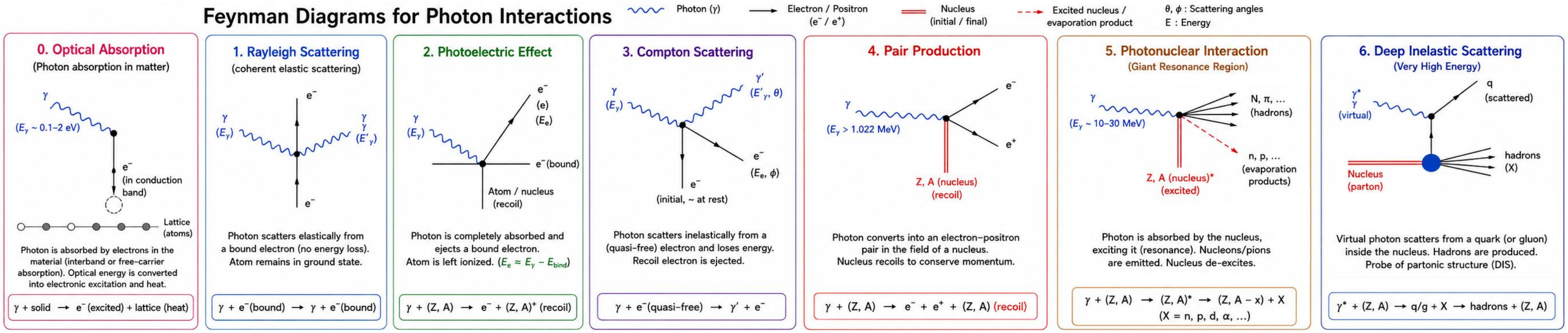
- Module impact estimation by incremental parametric sweep
- Module compute cost (C_m) estimation by wall time, CPU time, and memory cost
- Importance score (η_m) per module
- Module-by-module reduction opportunities

$$I_m = \frac{\left| y_{device}^{ref} - y_{device}^{(-m)} \right|}{y_{device}^{ref}}, \qquad \eta_m = \frac{I_m}{C_m}$$



E_{dep}	Energy deposition by radiation (Module 1)	ρ	Space-charge density (Module 2)	R	Recombination rate. Depends on C_j, n, p, T. (Module 5)
$C_j(\boldsymbol{r}, t)$	Concentration of defect species j (Module 3)	ϕ	Electric potential (Module 2)	n / p	Electron / hole concentration (Module 5)
D_j	Diffusivity of defect species j (Module 3)	$\boldsymbol{E}(\boldsymbol{r}, t)$	Electric field (Module 2)	J_n / J_p	Electron / hole current density (Module 5)
k_{ann}	Annealing rate for defect species j (Module 3, 4)	$\mu_n(T, C_j)$ / $\mu_p(T, C_j)$	Electron/hole mobility (Module 5)	y_{device}	Device response to radiation

Module 1: From Particle Interactions to Energy Deposition in Silicon using Geant4



$E_\gamma \approx [0.1\ eV, 6\ eV]$ | $E_\gamma \approx [1\ eV, 100\ keV]$ | $E_\gamma \approx [1\ eV, 500\ keV]$ | $E_\gamma \approx [10\ keV, 10\ MeV]$ | $E_\gamma \geq 1.022\ MeV$ | $E_\gamma \approx [5\ MeV, 3\ GeV]$ | $E_\gamma \geq GeV\ scale$

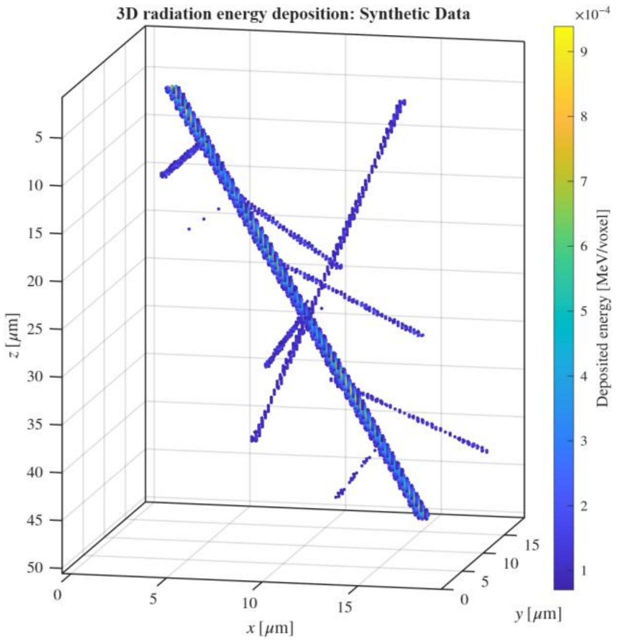
Geant4 Physics Library Used: FTFP_BERT

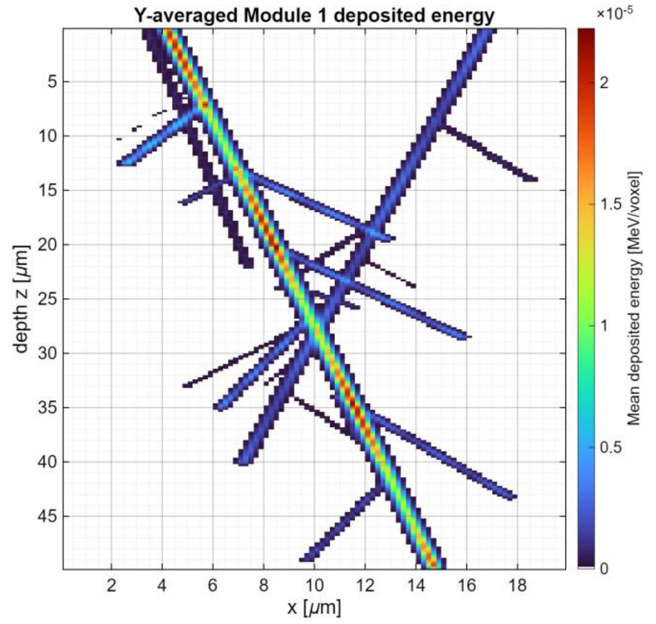
- Combines EM interactions with hadronic nuclear interaction models
- Provides event-by-event Monte Carlo outputs:
  - scattering, ionization, nuclear reactions, secondary particle production, deposited energy, and interaction histories.
- FTFP (Fritiof Precompound) is the high-energy model:
  - Handles violent nuclear collisions
  - Incoming particles strongly disrupts the nucleus
  - Produces secondary particles and leaves an excited leftover nucleus that must cool down.
- BERT (Bertini) is the lower-energy model:
  - Treats the nucleus as a crowded cluster of protons and neutrons
  - Incoming particles knocks into them one-by-one in a chain of collisions before particles escape.

Initial plan for Geant4 Simulations

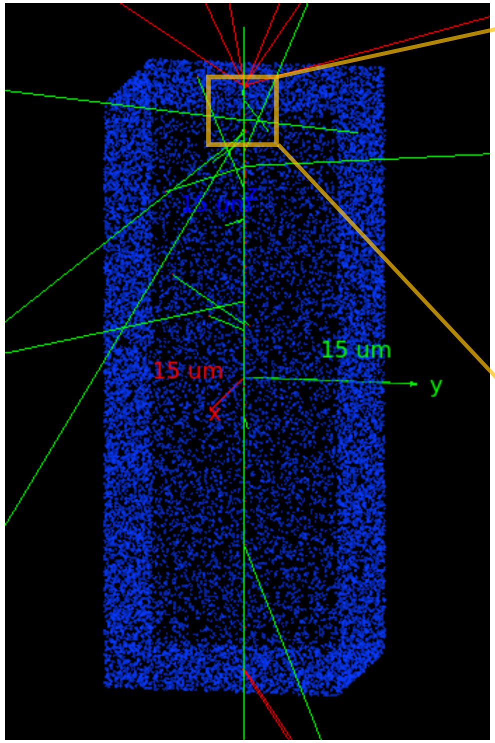
- Parametric sweep over $E_\gamma(1-10\ keV)$, intensity of the beam, and shielding design parameters.

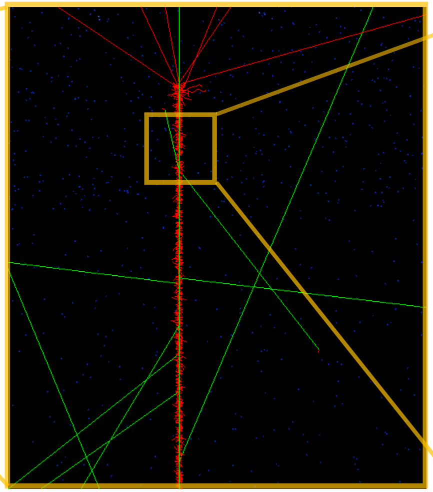
Expected Result for $5\ keV$ photon strikes





Module 1: Simulation of Particle Transport and Shielding Design with Geant4





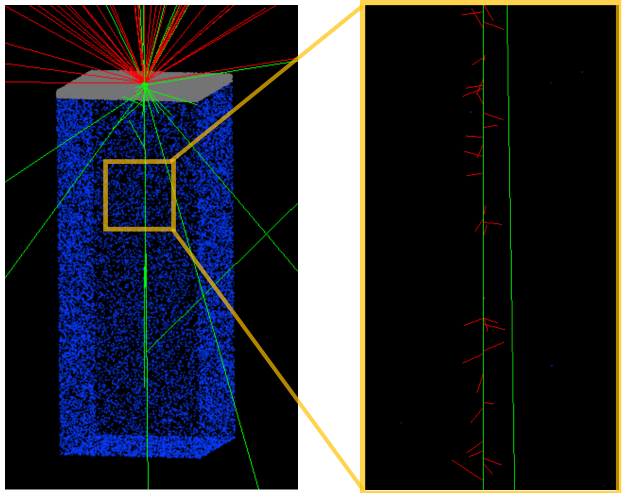
Base Run
- Silicon geometry: 20 um x 20 um x 50 um
- Photon energy: 5 keV
- Primary particle type: photons (green)
- Outcoming electrons (red)
- Flux: 50 events with 100 photons per event

Dominant Physics
- Photoelectric effect – dominant interaction
  - $\gamma + Si \rightarrow e^{-}_{photo} + Si^{+}$
  - A 5 keV photon is absorbed by a bound electron in silicon.
  - The photon disappears and a electron is emitted.
- Electron ionization – secondary electron production
  - $e^{-}_{photo} + Si \rightarrow e^{-} + e^{-}_{second} + Si^{+}$
  - Outcoming electrons from photoelectric absorption further collides and ionizes the silicon.
- Rayleigh scattering – elastic photon scattering
  - $\gamma + Si \rightarrow \gamma + Si$
  - Photons change direction but deposits little to no energy
- Analysis in progress

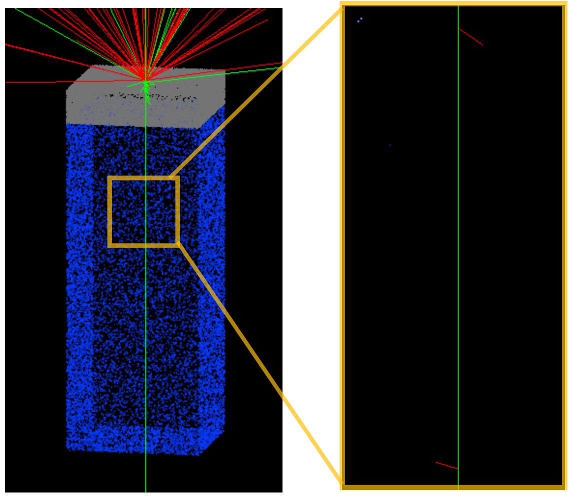
Shielding Design Implementation
- Tungsten for 5 keV photons
  - High Z material
  - More usage than lead for microelectronics (contacts, vias, plugs, and etc)
  - Expected attenuation length $\lambda \approx 0.94\ \mu m$
  - Transmission through tungsten $T \approx e^{-\frac{t}{\lambda}}$

Tungsten thickness	Approx. photon transmission	Out of 5,000 photons
$1\ \mu m$	$\sim 34\%$	~1,700 photons transmitted
$2\ \mu m$	$\sim 12\%$	~600 photons transmitted
$3\ \mu m$	$\sim 4\%$	~200 photons transmitted
$5\ \mu m$	$< 1\%$	~25 photons transmitted

Tungsten $1 \mu m$



Tungsten $5 \mu m$

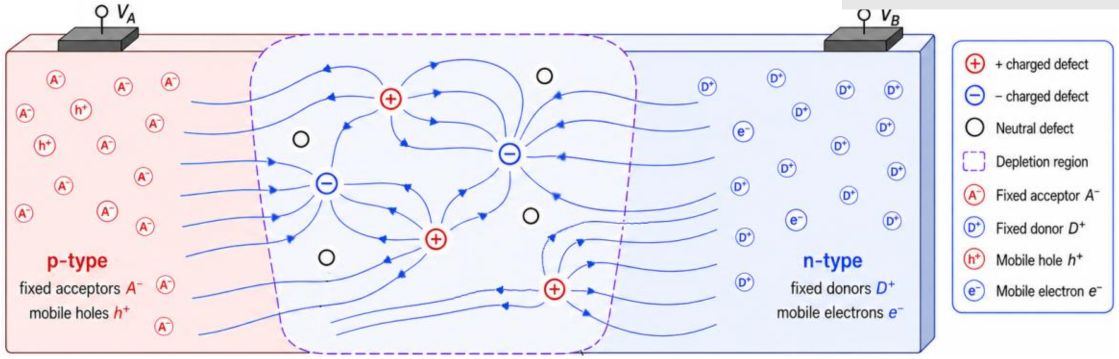


FEM: Governing Field Equations for Electrostatics, Defects, Heat, and Charge Carriers

- Module 2: Electric field perturbation from charged defects

$$\underbrace{\nabla \cdot (\epsilon_{Si} \nabla \phi)}_{\text{field spread in Si}} = \underbrace{-q(p - n + N_D^+ - N_A^-) + \rho_{def}}_{\text{charge source}}$$

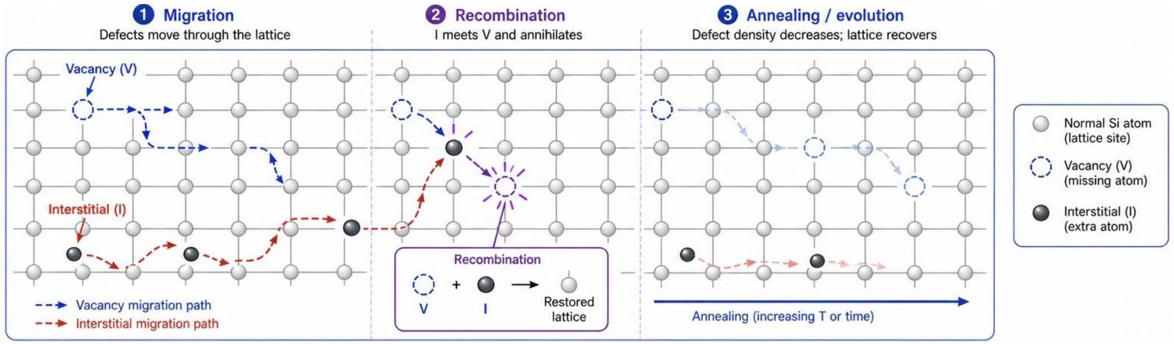
ϕ	electric potential
ϵ_{Si}	permittivity of silicon
n / p	electron / hole concentration
N_D^+ / N_A^-	donor/acceptor
ρ_{def}	defect charge density



- Module 3: Defect diffusion and annealing

$$\underbrace{\frac{\partial C}{\partial t}}_{\text{defect buildup}} = \underbrace{\nabla \cdot (D \nabla C) - k_{ann} C + S}_{\text{defect spread – defect removal}}$$

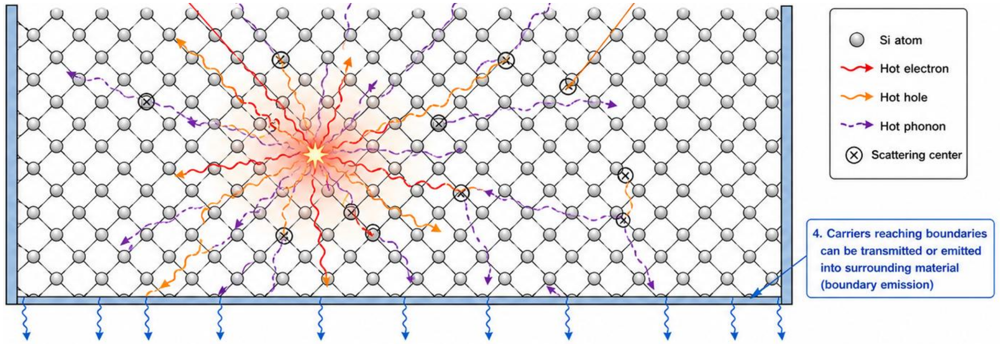
C	defect concentration
D	defect diffusivity
k_{ann}	1st order annealing coeff.
S	optional defect source



- Module 4: Ballistic-diffusive thermal transport

$$\underbrace{\tau C \frac{\partial^2 T}{\partial t^2} + C \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T)}_{\text{thermal response + heat spreading}} = \underbrace{Q + \tau \frac{\partial Q}{\partial t} - \nabla \cdot \boldsymbol{q_b}}_{\text{rad. heating + ballistic heat flow}}$$

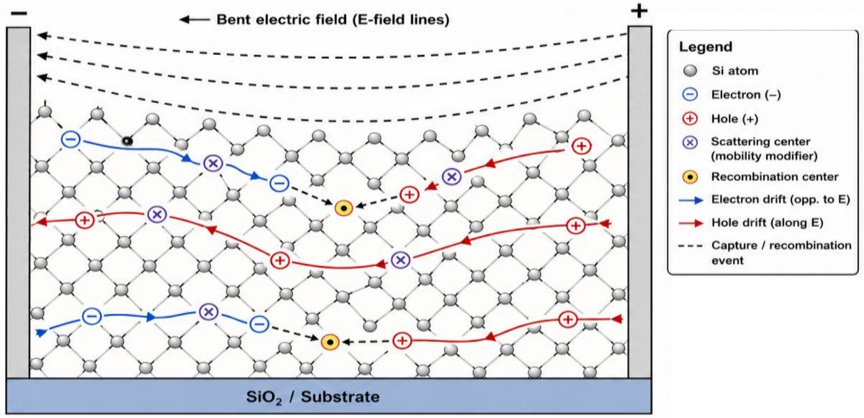
C	heat capacity
K	thermal conductivity
Q	heat generation rate density $[W/m^3]$
τ	thermal relaxation time
$\boldsymbol{q_b}$	ballistic heat flux vector



- Module 5: Charge carrier transport with recombination

$$\underbrace{\frac{\partial c}{\partial t}}_{e^-/h^+ \text{ build up}} = \underbrace{\nabla_{xy} \cdot \left(D_c \nabla_{xy} c + \sigma_c \mu_c c \boldsymbol{E}\right) + G - R}_{\text{diffusion + } \boldsymbol{E} \text{ field drift + generation – recomb.}}$$

c	charge carrier concentration: electron (e^-) or hole (h^+)
D_c	electron diffusivity when e^- hole diffusivity for h^+
σ_c	$+1$ for e^- and -1 for h^+
μ_c	electron/hole mobility
$\boldsymbol{E}$	electric field
G	e^- and h^+ pair generation rate
R	recombination rate



Conclusion: From Full Physics to Efficient Prediction

- Key Takeaways
  - ❑ Developed a modularized simulation roadmap from radiation energy deposition to silicon device degradation.
  - ❑ Modularized development reduces debugging effort.
  - ❑ Impacts of shielding/mitigation methods characterized separately per each physics phenomena.
  - ❑ Start from validated baseline model and increase physical complexity incrementally.

- Next Steps
  - ❑ Continue FEM code development for Module 2-5
  - ❑ Coupling sequence improvement: Module 6
  - ❑ Decoupled model and impact score estimation
  - ❑ Extrapolation back to full-scale interpretation
  - ❑ Uncertainty quantification